

EXPERIMENT - 17

COUNTING SEMAPHORE IN SYSTEMC

CODE:

```
#include <systemc.h>

SC_MODULE(counting_semaphore){
    int sem;

    void task()  {
        while (true){
            wait(1, SC_SEC);
            if (sem > 0)      {
                sem--;
                cout << "Task accessing resource at "
                    << sc_time_stamp()
                    << " | Remaining: " << sem
                    << endl;
                wait(2, SC_SEC);
                sem++;
                cout << "Task released resource at "
                    << sc_time_stamp()
                    << " | Available: " << sem
                    << endl;
            }
        }
    }

    SC_CTOR(counting_semaphore)  {
        sem = 2;
    }
}
```

```

        SC_THREAD(task);

        SC_THREAD(task);

        SC_THREAD(task);

    }

};

int sc_main(int argc, char *argv[])

{

    counting_semaphore obj("Counting");

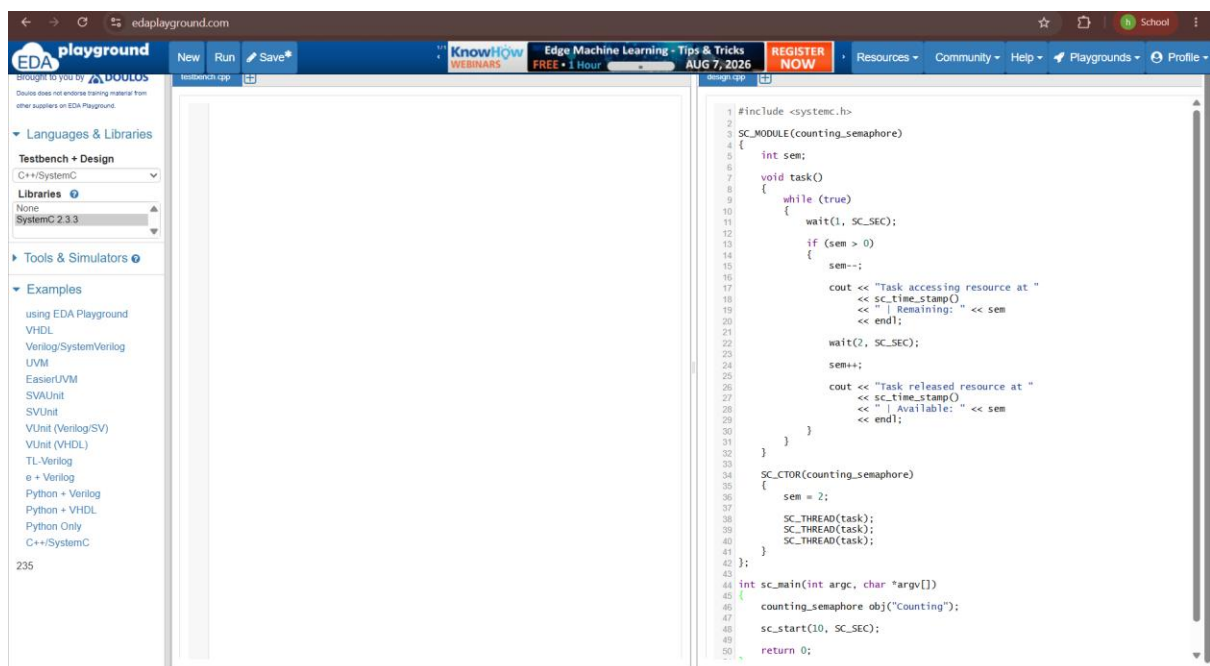
    sc_start(10, SC_SEC);

    return 0;

}

```

SIMULATION:



OUTPUT:

```
Log Share
[2026-08-11 16:17:44 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: ../../../../src/sysc/kernel/sc_object_manager.cpp:153

Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: ../../../../src/sysc/kernel/sc_object_manager.cpp:153
Task accessing resource at 1 s | Remaining: 1
Task accessing resource at 1 s | Remaining: 0
Task released resource at 3 s | Available: 1
Task accessing resource at 3 s | Remaining: 0
Task released resource at 3 s | Available: 1
Task accessing resource at 4 s | Remaining: 0
Task released resource at 5 s | Available: 1
Task accessing resource at 5 s | Remaining: 0
Task released resource at 6 s | Available: 1
Task accessing resource at 6 s | Remaining: 0
Task released resource at 7 s | Available: 1
Task accessing resource at 7 s | Remaining: 0
Task released resource at 8 s | Available: 1
Task accessing resource at 8 s | Remaining: 0
Task released resource at 9 s | Available: 1
Task accessing resource at 9 s | Remaining: 0

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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Done
```